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7. Can collisions be completely eliminated by using collision avoidance protocols like CSMA/CA?

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6. CSMA/CD requires that members of the network be able to detect if their transmissions interfere with that of others. In a wireless network, one member may not be able to hear all the other members or a member's transmissions may drown out those of a more distant member.

7. Wait for bus to become silent;

start transmitting;

if collision occurs, wait before trying again.

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How does a hub differ from a repeater?

How does a router differ from such devices as repeaters, bridges, and switches?

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How does a router differ from such devices as repeaters, bridges, and switches?

A hub connects individual computers to form a bus network, whereas a repeater connects two bus networks to form a larger network.

A router connects two networks to form an internet, whereas the other devices connect networks to form a larger network.

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What is the distinction between N-unicast and multicast?

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N-unicast forces the server to send multiple copies of each message (one to each client), whereas multicast allows the server to send only one copy, which is distributed to each of the clients.

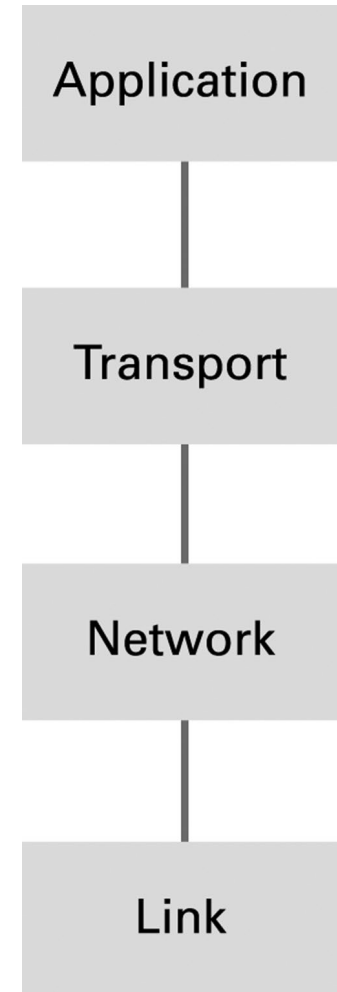
# 标题

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\*40. List the four layers in the Internet software hierarchy and identify a task performed by each layer.

# Internet Software Layers

- **Application:** Constructs message with address
- **Transport:** Chops message into packets
- **Network:** Handles routing through the Internet
- **Link:** Handles actual transmission of packets
  - if that network is an Ethernet, the link layer applies CSMA/CD. If the network is a WiFi network, the link layer applies CSMA/CA.





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**\*43.** In what way could TCP be considered a better protocol for implementing the transport layer than UDP? In what way could UDP be considered better than TCP?

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43. TCP actually confirms that the entire message made it to the destination, whereas UDP does not. However, UDP is more efficient.

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- \*45.** At what layer in the TCP/IP protocol hierarchy could a firewall be placed to filter incoming traffic by means of
- a. Message content
  - b. Source address
  - c. Type of application

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45. a. At the application layer
- b. At the network layer
  - c. At the transport layer